B.Tech Project – Weekly Progress Record

Student Name: Revelli Tharun **Admission No.: 23JE0807**

Date	Work Done	Student Signature	Supervisor signature
07/08/2026	Project allocation and literature survey on mobile social engineering attacks. Analyzed smishing and vishing evasion techniques, adversarial Cyrillic homoglyphs, and zero-width spaces.		
14/08/2026	Reviewed existing mobile security tools and SMS keyword blockers. Identified core research gaps: high battery/CPU overhead, cloud privacy violations on personal chats, and preview truncation.		
21/08/2026	Formulated system functional and non-functional requirements. Defined the Mobile Security Trilemma (sub-millisecond triage, zero-knowledge interpersonal privacy, and high ML accuracy).		
28/08/2026	Designed two-stage Edge-Cloud Sentinel architecture: Stage 1 on-device edge triage filter and Stage 2 deep cloud transformer NLP forensics. Selected Python FastAPI and Android Kotlin stacks.		
04/09/2026	Designed and dimensioned space-efficient Bloom filter using MurmurHash3 (k=7 hash functions) to hold 50,000 malicious phishing domains within an 85 KB memory footprint for sub-0.05 ms lookup.		
11/09/2026	Curated and preprocessed empirical benchmark dataset of 1,124 real-world communication payloads. Formulated feature extraction pipeline and evaluated pre-trained transformer architectures.		
18/09/2026	Trained and calibrated DeBERTa-v3 and MiniLM models with disentangled attention. Built soft-voting ensemble combining NLP confidence with domain forensics (spoofed brands, risky TLDs).		
25/09/2026	Developed asynchronous FastAPI REST microservice (/api/v1/detect, /health). Integrated OpenAI Whisper acoustic transcription pipeline for speech-to-text processing of voice call scam lures.		
01/10/2026	Built responsive dark-themed SOC web dashboard (localhost:8000) for real-time risk gauges, threat category breakdown radars, token attributions, and live visual defense demonstrations.		
09/10/2026	Developed native Android client (com.scamguard.app) in Kotlin. Implemented SmsReceiver for telephony broadcasts and NotificationListenerService for passive status-bar message interception.		
16/10/2026	Implemented Dual-Ingestion paradigm: Android Share Sheet (ACTION_SEND) for authoritative verification and one-tap Clipboard Paste-and-Scan. Integrated local SQLite/Prefs scan history.		
23/10/2026	Faculty review rebuttal: Engineered automated truncation detection for clipped previews ('...'), and active e-commerce noise blacklist (Amazon, Meesho, Swiggy) achieving 90.3% local request reduction.		
06/11/2026	Implemented On-Screen Chat Accessibility fallback for Notifications-OFF scenario and Contact Whitelist Privacy Shield. Validated 100-notification concurrent burst queue with 0 ANRs.		
12/11/2026	Conducted physical hardware benchmarking on OPPO K13 5G (Android 16, 0.24-0.56 ms latency). Finalized thesis report, 11-slide presentation deck with IIT ISM logo, and progress documentation.		